Dual-fusion Models for AI Image Attribution

Stanford CS229 Project

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1 Introduction

The increasingly high performance of generative models has made AI-generated images incredibly realistic and difficult for humans to identify, raising serious concerns about misinformation, impersonation, and trust. In light of this, the field of AI forensics has made substantial progress on synthetic image detection. Early evidence from Wang et al. (2020) showed that classifier models can take advantage of visual artifacts created by image generators to identify synthetic images with a high degree of accuracy. Subsequent work has has achieved a high level of robustness to images form unseen models (Ojha et al., 2023). That being said, identifying the specific model that produced an image (i.e. image attribution) remains relatively less mature. Existing attribution models can perform well under controlled settings but often degrade under common post-processing transformations (e.g., compression, resizing), which are prevalent in real-world scenarios (Yang et al., 2022; Bongini et al., 2025).

In this project, we attempt to address this gap by designing a dual-flow neural architecture that incorporates both spatial (RGB) and frequency features to perform image attribution. Building on recent evidence from Ricco et al. (2025); Rozhbayani et al. (2025), we hypothesize that combining these complementary signals will improve robustness to image transformations while still achieving high accuracy by leveraging generator frequency fingerprints when they are available. Specifically, the input to our model is an AI-generated image drawn from one of ten generators (e.g., Stable Diffusion 3.5, DALL-E 2, etc). We process the RGB image in one branch and a 2D Fast Fourier Transform (FFT) representation in a second branch, then fuse the learned features for final classification. The output is a multiclass prediction of the source generator.

2 Related Work

Recent advances in attribution models, particularly in frequency fingerprinting, patch-level learning, and transformer-based architectures, directly informed our approach. A central result in digital image forensics is that generative models leave unique fingerprints in the frequency domain of images they generate (Marra et al., 2019; Yu et al., 2019). Recent approaches exploit this, combining spatial and spectral image features to perform model attribution. Ricco et al. (2025) uses RGB and discrete fourier transform (DFT) radial signatures, and Rozhbayani et al. (2025) combines RGB and FFT branches in a dual-stream architecture, both achieving strong performance. These results show that frequency-domain patterns are a strong signal for generator identification.

Patch-level learning has also become an important strategy for image attribution. Bongini et al. (2025) demonstrates that CNNs that take random patches of an image as input achieve higher accuracy on plain images source attribution compared to models that take the entire image as input. This indicates that patch-based classifiers can better leverage the image artifacts created by image generators and, as such, are attractive for attribution settings where semantic content varies widely across prompts and classes.

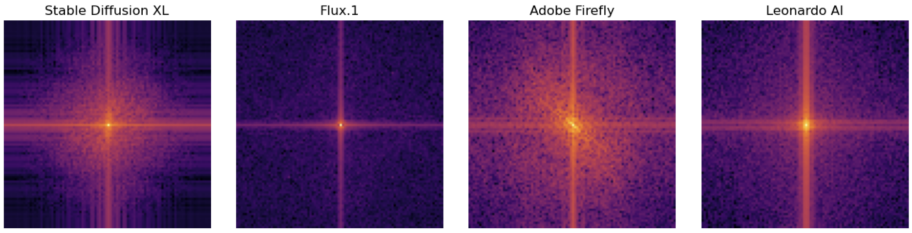
Finally, transformer-based visual models have been shown to be a strong alternative to CNNs for fake-image detection. Vision Transformers (ViTs) (Dosovitskiy et al., 2021) explicitly model long-range dependencies, differing from CNNs in ways consistent with stronger global feature use (Raghu et al., 2021). When applied to fake image detection tasks, vision transformers improved both generalization to unseen generators and robustness to processed images (Cozzolino et al., 2024). Together, these findings shows that transformer-based models can retain useful forensic signals for image attribution that are more robust to those of CNNs.

The WILD benchmark by Bongini et al. (2025) highlights the remaining gap: attribution methods that perform well on clean images degrade under realistic post-processing conditions that are likely to be encountered in the real world. Results indicate that ViT-based models are often more resilient than pure CNN baselines under these shifts, but their performance is still inferior to CNN models on plain images. Motivated by this limitation, our work targets a hybrid design that combines the robustness of ViT RGB processing with the accuracy provided by CNNs and frequency signals.

3 Dataset and Features

For the entirety of our project we used the WILD dataset, introduced by Bongini et al. (2025). This contains a closed set of 10,000 images, 1,000 from each 10 of the most popular commercial and open source text-to-image generators (Adobe Firefly, Dall-E 3, Flux.1, Flux.1.1 Pro, Freepik, Leonardo AI, Midjourney, Stable Diffusion 3.5, Stable Diffusion XL, and Starry AI). All of the images had randomized prompts within a given structure such as: <subject> with <facial features> wearing <clothing> <be> looking <looking direction> (in <expression(emotion)>) (<background>). This lead to 1024x1024 pixel images of faces that look like this:

In order to extract the images' frequency spectrum, we passed their RGB pixels through a 2D discrete Fourier transform then shifted and log-scaled the output, consistent with the methodology of Rozhbayani et al. (2025). We then split the data to have 7,000 images in our training set, 1,500 images in our validation set, and 1,500 images in our test set. We stratified this sampling by class, so there are 700 images from each model in the training set and 150 images from each model in the validation and test sets. We resized each array, such that each image now has dimension [224, 224, 3, 2]. Each of these arrays is normalized so that the RGB domain has means [0.485, 0.456, 0.406] and standard deviations [0.229, 0.224, 0.225] and the frequency domain has means [0, 0, 0] and standard deviations [1, 1, 1]. Some of the frequency domain arrays are displayed here:



To prevent overfitting, we randomly applied various transformations to our training images. The first step of this processing is to randomly "zero out" either the RGB or frequency array for certain images by setting the corresponding array to zeros. For the RGB part of the image, we randomly translated, flipped (horizontally and/or vertically), rotated (up to 15 degrees), blurred, and jittered with specified probabilities between 0.1 and 0.5. For the frequency part of the image we randomly added Gaussian noise and masked certain regions with probabilities 0.4 and 0.05. The goal of these transformations is to force our models to learn high-level differences between images rather than overfitting to granular features.

We used use 2 different pre-processing techniques in training our models. The first technique is to simply consider the images as-is, using the procedure outlined above. The second technique is to only train the model on a 96-by-96 pixel "patch" of the images. For this technique, we randomly selected a subarray of size [96, 96, 3] from each image and then applied a Fourier transform as well as all of the above transformations to that patch. At inference, we create "batches" of 50 such "patches" for the testing images, making the test set images have dimension [50, 96, 96, 3, 2]. The model outputs prediction probabilities for each of these 50 "patches," which are then aggregated to achieve a final prediction. This technique was used in an attempt for the model to generalize better to "non-face" images by focusing on frequency patterns from each model rather than specific facial features that may be unique to a certain model.

The WILD dataset also contains post-processed face images with three versions of the same images, step 1, step 2, and step 3. Each step progressively adds one random transformation from a set of ten, including

compression, random cropping, rotation, and contrast/brightness adjustment in order to simulate real-world conditions. We used this as an additional test set to evaluate generalization to distorted images.

We also generated 100 images each from Stable Diffusion 3.5 and Stable Diffusion XL, that use randomly generated prompts, not necessarily containing faces or people like the images in the WILD dataset. This was used as a check to see how well the model generalized to non-facial images. These images used prompts such as "A panoramic view of a megacity at night", "A documentary-style photo of a farmer in rural India", and "A vast desert with giant alien monoliths at sunset", very different and more reflective of the true generalizability of the models. Some of these images are shown here:

4 Methods

The primary model we propose is the Dual Stream Fusion model, depicted in Figure 1. The architecture consists of two parallel neural-network streams. The spatial (RGB) stream uses a Vision Transformer (ViT-B/16) pre-trained on ImageNet. To leverage ViT pretraining and reduce training time, we freeze the weights of this stream during training. The second stream processes the frequency domain using a ResNet-50 architecture. Unlike the ViT, the ResNet-50 is initialized without pre-trained weights and is trained from scratch on the frequency stream. The ViT model requires 224×224 inputs, which is why both streams are resized before being fed to the model.

To produce a final prediction, the feature vectors from both streams are fused. Let $f_{rgb} \in \mathbb{R}^{d_1}$ and $f_{freq} \in \mathbb{R}^{d_2}$ denote the extracted feature embeddings from the ViT and ResNet-50, respectively. We concatenate these vectors to form a combined representation, $z = [f_{rgb} \oplus f_{freq}]$. This fused vector is passed through a multi-layer perceptron (MLP) classification head consisting of a linear layer, a ReLU activation function, dropout regularization ($p = 0.5$), and a final linear layer that outputs raw class logits. Final class probabilities are computed with Softmax.

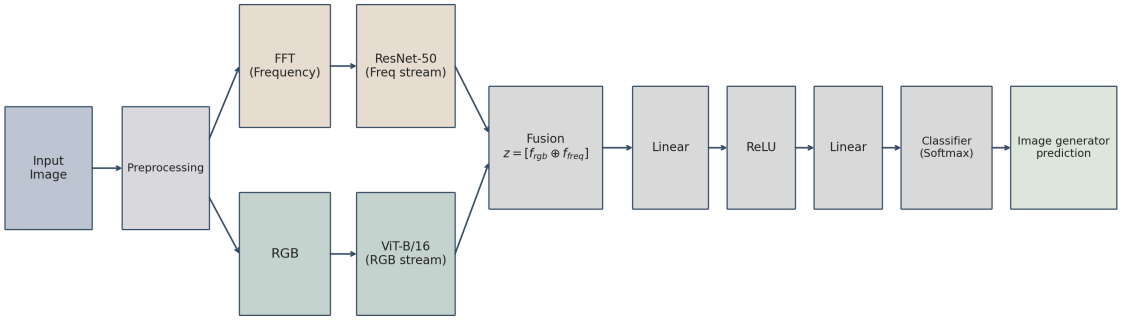


Figure 1: Dual Stream Fusion architecture.

This architecture is inspired by the model developed by Rozhbayani et al. (2025), with two key differences: (1) the frequency stream is passed through a ResNet-50 architecture and (2) the model is used for multiclass classification (image attribution) instead of binary classification (real-vs.-fake image detection). The first choice is motivated by previous work showing that CNNs are better suited to detect localized high-frequency patterns than transformer models (Wang et al., 2020; Dosovitskiy et al., 2021).

We also developed a variant of this architecture, Dual ResNet (Figure 2), which replaces the ViT branch with a second ResNet-50 and follows the patch-based pipeline. This design avoids resizing the full image, since resizing can weaken high-frequency generator fingerprints. Instead, each sample is represented using 96×96 RGB patches and their corresponding frequency representations. The RGB ResNet-50 uses weights pretrained on ImageNet, while the frequency branch ResNet-50 is trained from scratch. During training, we pass one randomly sampled 96×96 patch per image. During inference, we sample 50 patches per image, so each input is a 5D tensor: $(\text{Batch}, 50, 3, 96, 96)$. We then flatten the patch dimension into the batch dimension to process

all patches in parallel. For each patch, RGB and frequency features are concatenated and passed through the same MLP head used in Dual Fusion, producing per-patch logits. The final image-level prediction is obtained by averaging logits across all 50 patches before applying Softmax. Mathematically, the image-level logit vector L_{img} is defined as

$$L_{img} = \frac{1}{50} \sum_{k=1}^{50} \text{MLP} \left(\text{ResNet50}_{rgb}(p_k^{rgb}) \oplus \text{ResNet50}_{freq}(p_k^{freq}) \right)$$

where p_k denotes the k-th extracted patch.

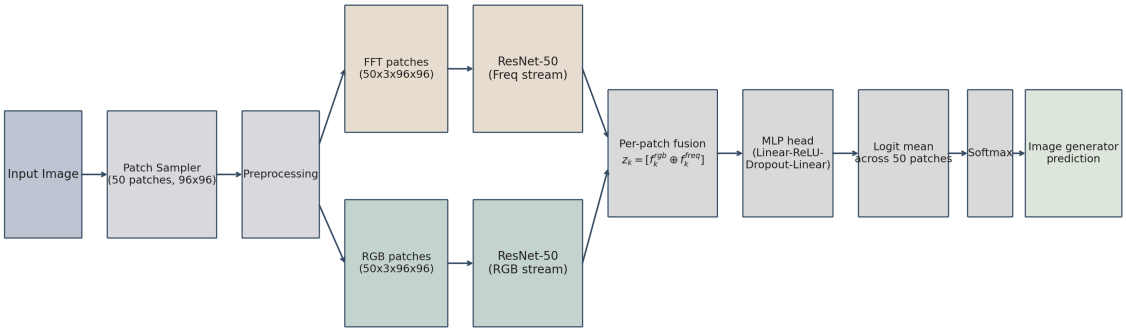


Figure 2: Dual ResNet patch-based architecture.

The Dual ResNet architecture follows a methodology similar to that of Bongini et al. (2025), who also used 50 96×96 RGB patches as input to a ResNet-50 model. However, their approach does not incorporate frequency features or a dual-stream architecture.

In addition to these two models, we trained and evaluated three baselines. The first is a ResNet-50 that follows the WILD methodology: it takes 50 96×96 RGB patches, averages the patch logits and applies Softmax for the final prediction. The second is a ViT baseline that also follows the WILD methodology, using a ViT model that splits each image into 16×16 patches within the forward pass and aggregates patch-level predictions to produce the final classification score. The third baseline is a ResNet-50 that takes the full RGB image as input, without patching. All three were initialized with weights pretrained on ImageNet.

5 Experiments and Results

We trained all 5 models on the training set, containing plain images from the WILD dataset. We utilized Cross Entropy loss as the loss function and AdamW as the optimizer. The models were trained for a maximum of 50 epochs with an early stopping condition. At every epoch, loss was calculated on the validation set, and training would come to a halt if the validation loss did not decrease by a minimum delta of 10^{-4} for 5 consecutive epochs. Finally, we utilized a batch size of 32 and a learning rate of 10^{-4}. The 10^{-4} learning rate increases weight preservation and is an established baseline for the AdamW optimizer for transfer learning (Yosinski et al., 2014; Kingma and Ba, 2015). Furthermore, the mini-batch size of 32 was selected as it has been used as a strong baseline for standard CNNs (Masters and Luschi, 2018; He et al., 2019). Model parameters were tuned utilizing the validation set as well.

All 5 models were evaluated on the test set (plain images), the entirety of post-processed set (step 1, 2 and 3), and the "Not Faces" dataset we curated. Table 1 displays balanced accuracy for each model across all data sets while table 2 displays ROC-AUC scores for each model across all data sets.

Both proposed models (Dual Fusion and Dual ResNet) performed strongly on the plain test split, which is from the same distribution as the training set, but degraded substantially as post-processing severity increased. For example, Dual Fusion balanced accuracy dropped from 95.47% (Plain) to 29.00% (Step 3), and Dual ResNet dropped from 100.00% to 37.34%. The same trend appears in ROC-AUC, showing that class separability declines under realistic distortions. This pattern motivated our hypothesis that the model may over-rely on frequency cues, which are more fragile to post-processing. If the model was over reliant on the frequency stream and the transformations applied to the images completely eliminated the generator fingerprints in the frequency domain, then we would expect a pattern like the one seen above.

Table 1: Model Balanced Accuracy across WILD Closed Set.

Model	Plain	Step 1	Step 2	Step 3	Not Faces
Dual Fusion	95.47%	57.30%	38.32%	29.00%	29.00%
Dual ResNet	100.00%	70.84%	50.52%	37.34%	37.24%
ResNet50(WILD)	99.80%	75.96%	59.76%	46.98%	46.28%
ViT (WILD)	95.20%	94.64%	90.40%	86.60%	86.60%
ResNet50	98.80%	95.64%	90.22%	87.12%	87.12%

Table 2: Model ROC-AUC scores across WILD Closed Set.

Model	Plain	Step 1	Step 2	Step 3	Not Faces
Dual Fusion	0.9986	0.9100	0.8017	0.7189	0.7189
Dual ResNet	1.0000	0.9564	0.8843	0.8113	0.8106
ResNet50(WILD)	1.0000	0.9598	0.9019	0.8340	0.8358
ViT (WILD)	0.9983	0.9989	0.9969	0.9944	0.9944
ResNet50	0.9998	0.9988	0.9958	0.9923	0.9923

To test this hypothesis, we employed a method to quantify the contribution of the RGB branch and the frequency branch to the predictions. Focusing on our Dual ResNet model, we calculated the SHAP values for each node in the last layer on the validation set. The final layer in our dual flow architecture contains 4096 nodes. Each of the inputs of the vector of final layer nodes (size 4096) is considered as a "player" in the final model prediction, and SHAP uses cooperative game theory to quantify the contribution of each "player" to the prediction. The first 2048 entries of this vector are the RGB domain information, and the remaining 2048 entries are the frequency domain information. We calculated the mean of the SHAP values for the RGB and frequency entries and deduced that the RGB domain accounts for 47.33% of prediction contribution and the frequency domain accounts for 52.67% of prediction contribution. In short, this means that our model places approximately equal weight on RGB and frequency information from the images, which doesn't show an meaningful overreliance.

Another notable result is that the single-stream ResNet50 baseline remained highly robust on transformed and non-face data. It outperformed ResNet50(WILD) and approached ViT-level robustness while using substantially fewer parameters (about 25M vs. 86M), suggesting that, in our setting, simpler full-image representations may generalize better than patch-based or dual-stream variants.

6 Conclusion

Both proposed architectures Dual ResNet and Dual Fusion achieved strong performance on plain WILD images, in line with methods from the recent literature. However, performance dropped sharply on post-processed and non-face datasets. This indicates that these architectures, at least as they stand, don't achieve the robustness we hoped against image processing and in-the-wild scenarios. The results also indicate that simpler models that processed the entire image generalized better under image transformations. ResNet50 and ViT maintained substantially higher balanced accuracy and ROC-AUC, with ResNet50 offering a strong robustness–efficiency trade-off (about 25M vs. 86M parameters). It might be worth exploring architectures that can downweight compromised features, so dual-stream architectures can benefit from the predictive strength of frequency-domain information without sacrificing robustness to common transformations. It would also be interesting to perform an interpretability analysis to uncover which aspects the ResNet50 model is capturing. Once we understand the components that are most important for image classification, it may allow us to create a model that is especially tailored for detecting and classifying AI-generated images.

It might be the case that the key to making a robust attribution model that is more generalizable may be training on the most diverse dataset possible. The scope of the WILD dataset is quite narrow and it is hard to extract information about the image-generator without overfitting to the types of faces they produce. If we had more time and resources, it could be interesting to build our own dataset for training, with completely randomized prompts to get more diverse images.

7 Contributions

- **Trent Bellinger**: Data pre-processing and model exploration
- **Joao Panicali**: Model development and training
- **Ian Milstein**: Error analysis, writeup

References

Pietro Bongini, Sara Mandelli, Andrea Montibeller, Mirko Casu, Orazio Pontorno, Claudio Vittorio Ragaglia, Luca Zanchetta, Mattia Aquilina, Taiba Majid Wani, Luca Guarnera, et al. 2025. Wild: a new in-the-wild image linkage dataset for synthetic image attribution. In *2025 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE.

Davide Cozzolino, Giovanni Poggi, Riccardo Corvi, Daniel Nielberg, and Luisa Verdoliva. 2024. Raising the bar of ai-generated image detection with clip. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 4356–4365.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*.

Tong He, Zhi Zhang, Hang Zhang, Zhongyue Zhang, Junyuan Xie, and Mu Li. 2019. Bag of tricks for image classification with convolutional neural networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 558–567.

Diederik P. Kingma and Jimmy Ba. 2015. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*.

Francesco Marra, Diego Gragnaniello, Davide Cozzolino, and Luisa Verdoliva. 2019. Do gans leave artificial fingerprints? In *2019 IEEE Conference on Multimedia Information Processing and Retrieval (MIPR)*, pages 506–511. IEEE.

Dominic Masters and Carlo Luschi. 2018. Revisiting small batch training for deep neural networks. *arXiv preprint arXiv:1804.07612*.

Utkarsh Ojha, Yuheng Li, and Yong Jae Lee. 2023. Towards universal fake image detectors that generalize across generative models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24480–24489.

Maithra Raghu, Thomas Unterthiner, Simon Kornblith, Chiyuan Zhang, and Alexey Dosovitskiy. 2021. Do vision transformers see like convolutional neural networks? *Advances in neural information processing systems*, 34:12116–12128.

Emanuele Ricco, Elia Onofri, Lorenzo Cima, Stefano Cresci, and Roberto Di Pietro. 2025. Prism: Phase-enhanced radial-based image signature mapping framework for fingerprinting ai-generated images. *arXiv preprint arXiv:2509.15270*.

Gona Rozhbayani, Amel Tuama, and Huda Hamza Abdulkhudhur. 2025. Fusing frequency and spatial transformers for robust detection of ai-generated images. *Ingenierie des Systemes d'Information*, 30(10):2719.

Sheng-Yu Wang, Oliver Wang, Richard Zhang, Andrew Owens, and Alexei A. Efros. 2020. Cnn-generated images are surprisingly easy to spot... for now. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8695–8704.

Tianyun Yang, Ziyao Huang, Juan Cao, Lei Li, and Xirong Li. 2022. Deepfake network architecture attribution. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 4662–4670.

Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson. 2014. How transferable are features in deep neural networks? In *Advances in Neural Information Processing Systems*, volume 27, pages 3320–3328.

Ning Yu, Larry Davis, and Mario Fritz. 2019. Attributing fake images to gans: Learning and analyzing gan fingerprints. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7556–7566.